Autonomous Mass-Aware Sorting System

#Team 8 Armstrong

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Do you like online-shopping?

But sometimes …

What I Ordered vs What I Got

What I Ordered vs What I Got

Challenges in the Return Process

- Mixed-weight items handled together
- Weight measurement and sorting are separated, reducing efficiency

- Improper stacking may cause damage (heavy items crushing lighter ones)

Our Idea: Weight-Aware Robotic Sorting System

- **Process Optimization**

  Integrate weight measurement and sorting into a single efficient workflow

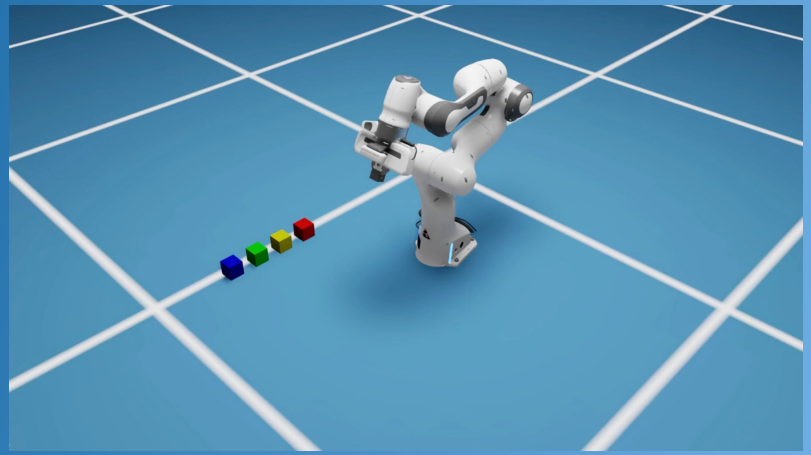
- **Control Strategy**

  Enable adaptive grasping by adjusting gripping force based on item weight

➢ Faster, safer, and more efficient return handling
➢ Applied to any conveyor-line system where weight detection is required

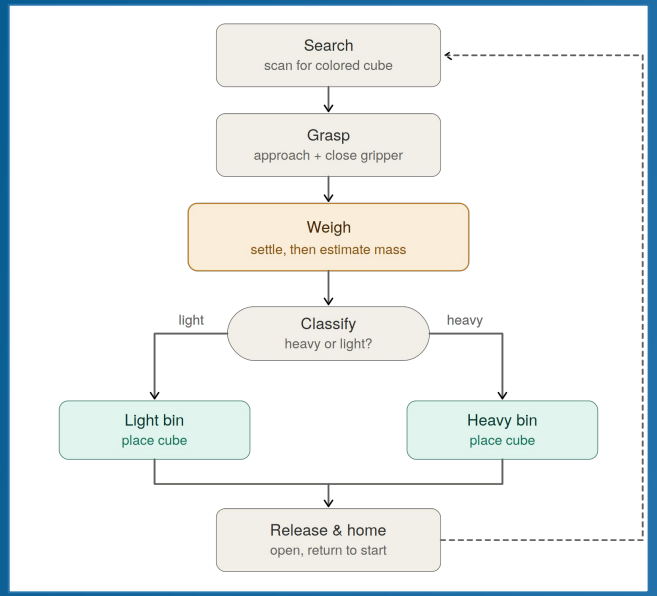
The simulation setup

- **Environment**: MuJoCo simulation (4-DOF arm, overhead camera).
- **Task**: Arm sees colored cubes, picks them up, sorts them by weight.
- **Catch**: No force sensor, no scale, weight must be inferred from control signals.



Source: NVIDIA Isaac Sim

How do you weigh without a scale?



Two ways to estimate mass:

PID-Error Method

Infer mass from how much the controller struggles to hold position under load.

- Pro: Very simple to implement
- Con: Gain-dependent, Pose-dependent

Inverse-Dynamics Residual Method

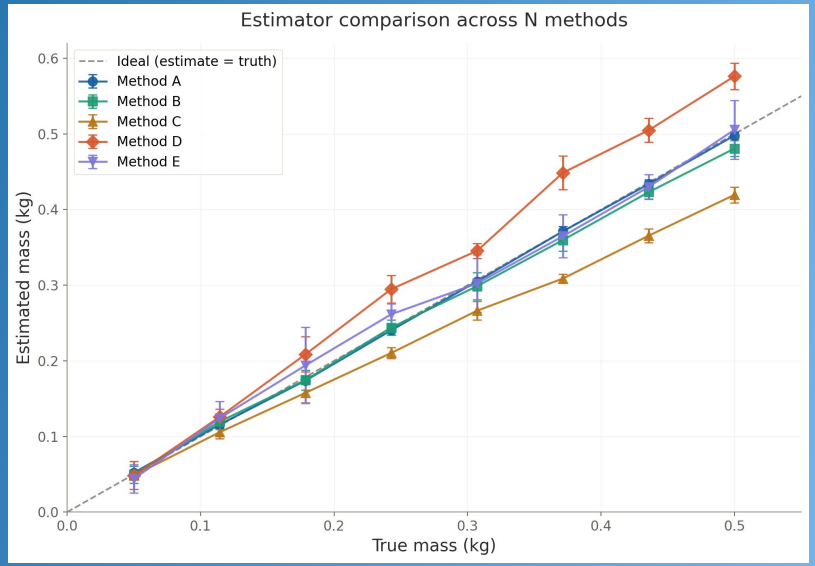
Infer mass from the torque left over after subtracting the arm's known gravity contribution.

- Pro: Gain-independent, Pose-independent
- Con: More complicated math and simulation setup

A Case Study: Comparing estimators

- **Question:**
  - **Accuracy, robustness, complexity: Which estimator wins which tradeoff?**
- **Scope:**
  - **Baseline: 2 estimators**
  - **Extended: +2 estimators**

Mock estimator-comparison results:



Roles & Gantt Chart

Phase	Task	Start	End	Primary owner	Support / controller contribution	W1 4/20	W2 4/27	W3 5/4	W4 5/11	W5 5/18	W6 5/25	W7 6/1	W8 6/8
Proposal	Finalize project scope, objectives, deliverables, duties, software/resource list, and proposal slides	4/19	4/20	All	All present part of proposal	■							
Modeling	Literature review + define control requirements/performance metrics + explicit assumptions	4/20	4/26	Lisha Yuan	All review metrics and assumptions	■							
Modeling	Derive FK/IK, workspace limits, camera-to-world transform, and pick/place coordinates	4/20	5/3	Michael Xiong	Kinematic for Control	■	■						
Modeling	DH table + Jacobian Matrix + Equations of motion	4/20	5/10	Han Wu	Dynamics for simulation	■	■	■					
Visualization	Select 3D robot model	4/20	5/3	Yonghao Huo	Animation and CAD	■	■						
Perception	Camera test, visual calibration	4/27	5/10	Han Wu & Michael Xiong	Sensor feedback testing		■	■					
Control	Baseline closed-loop joint-space PID tracking with setpoint hold and pick/place primitives	4/27	5/10	Han Wu & Michael Xiong	Tune gains on assigned poses with gravitational factor		■	■					
Integration	Trajectory integration: recognize -> approach -> grasp -> move -> sort -> drop -> home	5/4	5/17	Yonghao Huo & Lisha Yuan	Select the best trajectory			■	■				
Integration	Animation test + Supplementary CAD	5/4	5/14	Yonghao Huo & Lisha Yuan	Test liability and stability of visualization			■	■				
Control	Controller: PID vs gravity-comp/inverse-dynamics variant	5/11	5/31	Michael Xiong	Complete PID controllerfirst, and generate IDV controller if have time				■	■	■		
Data	Collect scenario data: end-effector position error, and joint angle, velocity, force error.	5/11	6/7	Lisha Yuan	Repeated runs and collect errors				■	■	■	■	
Progress	Prepare progress report: literature slide, methods to date, early results, challenges, updated timeline	5/11	5/14	All	All present some aspect				■				
Integration	Task simulation: robust pick-and-sort across target colors with logging/error plots enabled	5/18	6/7	Han Wu & Michael Xiong	Optimize code and verify performance metrics					■	■	■	
Data	Gain tuning, error assessment, robustness limits, and controller recommendation	5/25	6/7	Han Wu & Lisha Yuan	All compare controllers / justify recommendation						■	■	
Visualization	Capture final animation and figure sets	6/1	6/10	Yonghao Huo	3D Animation							■	■
Final	Final presentation	6/8	6/12	All	All rehearse technical explanation								■
Final	Final report writing: literature, methods, results, discussion, future work, appendix, code packaging	6/8	6/14	All	Sections authored by primary leads, merged by team								■

Roles & Gantt Chart

Member	Primary lead	Discription
Michael Xiong	Control code	Kinematic model, PID, controller comparison, and joint/gain tuning
Yonghao Huo	3D visualization	Trajectory and Animation
Han Wu	Simulation & integration	Dynamic model, integration tests and and joint/gain tuning
Lisha Yuan	Perception & data	Metrics, logging, robustness testing, and error study

Q&A

Thank You!